4 **(a)** Explain, in terms of the momentum of particles, how a gas exerts a pressure.

.....

.....

.....

..... [3]

(b) The temperature of a sample of gas is increased at constant volume.

State and explain any change in the pressure of the gas.

.....

.....

.....

..... [2]

(c) Another sample of gas is in a sealed container of volume 170 cm^3 and exerts a pressure of 9.0×10^4 Pa. The volume of the container decreases by 70 cm^3 at constant temperature.

Calculate the new pressure of the gas.

pressure = ... [3]

[Total: 8]